

Richard Xu

647-764-0174 | richardxu11111@gmail.com | GitHub | Portfolio | LinkedIn

EDUCATION

Western University

London, ON

Bachelor of Engineering Science (BESc), Software Engineering — CGPA: 3.9/4.0

Sept. 2024 – Apr. 2028

Ivey Business School, Honours Business Administration (HBA)

Sept. 2027 – April 2029

Coursework: Data Structures & Algorithms, Software Design, Software Construction, Scripting Language Fundamentals

EXPERIENCE

Forward Deployed Software Engineer

May 2026 – Present

Magical (Series A Startup)

Toronto, ON

- Developed production healthcare automations using **TypeScript, Temporal, LLM agents, Playwright, REST APIs, and Redis/Upstash** across EHR, CRM, Gmail, and Slack systems
- Reduced CSV export latency by **20× (5.5 minutes to 17 seconds)** by replacing nearly 13,000 per-record Redis requests with batched **HMGET** pipelines
- Built Slack-based run observability and defensive API handling, identifying incomplete executions and recovering malformed CRM payloads across 150 production runs
- Eliminated a **6.8% reporting reconciliation gap** across 1,600+ claims through cohort-based deduplication and validation

Software Engineering Intern

Jan. 2026 – April. 2026

Autumn (Startup)

New York, NY

- Built MVP for a full-stack platform to automate pricing and workflow processes for funeral service providers, simplifying complex and manual operations
- Developed backend logic and APIs using **Strapi (Node.js)** and **PostgreSQL** capturing data for over **15,000** funeral houses in the U.S
- Built a responsive frontend using **Next.js** to enable intuitive user interaction and workflow management

AI/ML Research Intern

May 2025 – Aug 2025

Western University

London, ON

- Built an intelligent Wi-Fi optimization platform that dynamically switched access points based on real-time network conditions to improve connectivity stability
- Designed and trained a **PPO reinforcement learning agent** using **Stable-Baselines3** and **Gymnasium**, with a custom environment optimized for latency, jitter, and packet loss
- Engineered a **Flask + React** experimentation platform and benchmarked the agent against static and random strategies across **50+ reproducible episodes per run**

PROJECTS

LLM Gateway | *TypeScript, Node.js, Hono, React, SQLite, WebSockets, SSE*

Sept. 2026

- Built an authenticating reverse proxy for the OpenAI API that streams SSE responses and ships request logs to a React dashboard with live WebSocket updates and time-series analytics

Slack Digest AI | *Javascript, Node.js, Express, Slack API, REST APIs*

April 2026

- Built a Node.js/Express service that pulls messages from Slack channels through the Slack API, groups them by thread, summarizes each with an LLM, and serves the result to a frontend as a daily digest
- Processed digests in under **10s**, batching threads into a single prompt to cut **LLM calls 85%**

eBay Deal Finder | *Python, Flask, React, eBay API, PostgreSQL, Render*

July 2025 – Aug. 2025

- Built a full-stack deal search on Flask and React that authenticates against the eBay API in sandbox and production, normalizes listing responses, and ranks results by price and rating
- Searched **100 live listings** per query and returned ranked results in **under 900ms**

TECHNICAL SKILLS

Languages: Python, Java, Javascript, HTML/CSS, C#, Typescript

AI / ML: LLMs (API-based), Prompt Engineering, RAG, Agentic Systems, Reinforcement Learning

Frameworks: React, Next.js, Flask, Django, FastAPI, Strapi (Node.js), .NET, Unity, Playwright

Libraries/Databases: pandas, NumPy, Matplotlib, PostgreSQL, Redis/Upstash